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PALAEOMAGNETIC INVESTIGATIONS IN THE HERCYNIAN MOUNTAIN BELT OF CENTRAL EUROPE *

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ABSTRACT

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The palaeomagnetism of sediments and volcanics of Middle Devonian to Early Carboniferous age has been studied in the Hercynian realm of the Vosges, Harz and Franconian Forest (France and Germany). After appropriate cleaning, characteristic directions of natural remanent magnetization (NRM) have been determined, which are considered to be of pretectonic origin according to rock magnetic and structural geological evidence. A mean palaeopole for the Devonian rocks in the areas investigated was determined at −29.7°N and 9.0°E. The site latitude derived from the Devonian rocks agrees very well with that observed in Great Britain and Britanny. The significant difference in longitude between our Devonian poles and the other European data may be due to an anticlockwise rotation of central Europe with respect to the northern Hercynian Foreland. The Carboniferous rocks yield a mean palaeopole at −41.7°N and 340.4°E, matching the Carboniferous part of the European polar wander path. A fold test within Carboniferous greywackes from the Harz mountains proves a pre-folding origin of the NRM of these rocks. However, this stable magnetization direction indicates local or regional tectonic displacement. Our results do not support any major latitudinal motions of continental segments in central Europe during the Hercynian orogeny. The Palaeozoic palaeomagnetic data from the Vosges, Franconian Forest, Harz, Britanny and Central Massif can be used to interpret the Hercynides in terms of a large scale shear belt.

INTRODUCTION

The application of plate tectonic concepts in the Hercynian domain of western and central Europe has been a subject of considerable debate for the last 10 years (Burrett, 1972; Floyd, 1972; Nicolas, 1972; Johnson, 1973, 1974; Dewey and Burke, 1973; Riding, 1974; Lorenz, 1976). Typical collisional features such as ophiolites, blueschists, melanges, indicating a high pressure and low-temperature regime are absent in the Hercynian orogen of Europe (Behr, 1978). Shape, width, tectonic style

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and zonation as well as rock types of the mountain belt do not allow for a simple comparison with well-known younger collision orogens such as the Alps, the Andes or the Himalayas (Zwart, 1967).

In view of these difficulties other evolutionary models have been proposed. Zwart (1967), Krebs and Wachendorf (1973) and Ager (1975) ascribed the Hercynian orogeny to a process of ensialic tectogenesis. They reject any large-scale plate motion in Europe during Hercynian times and deduce that the orogeny was caused by a significant thermal event, which would account for the widespread occurrence of Hercynian granites. Behr (1978), Weber (1978) and Behr and Weber (1980) attempted to explain the structural style of the Hercynides, particularly the crustal thickening (Giese, 1978; Kenolty et al., 1981) by proposing a subfluence model (i.e. upper mantle convection would produce laminar flow in the lower crust and underthrusting of discrete continental blocks). With this model opening and closure of a Palaeozoic ocean (Morel and Irving's (1978) Proterozoic Atlantic Two) is not required.

The present controversy about the different models for the Hercynian orogeny may be clarified with the help of geophysical methods, especially palaeomagnetism. Plate tectonic models require rotational movements of crustal units. Palaeomagnetic investigations which refer to a network connected to the earth's rotational axis, may prove or disprove the occurrence of such rotations in the Palaeozoic before and

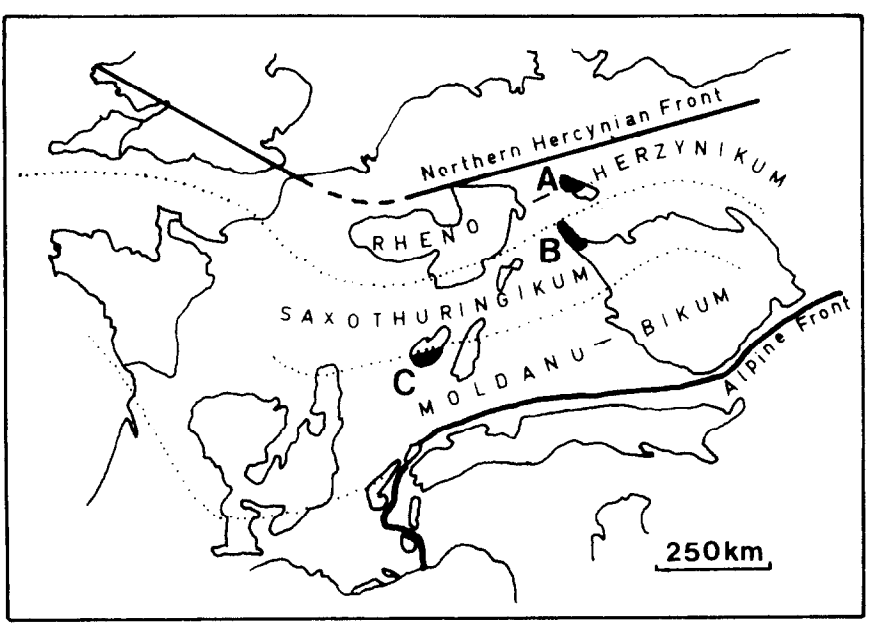


Fig. 1. Schematic map of the main outcrops of Palaeozoic rocks in central Europe and the zonation of the Hercynian mountain belt (Kossmat, 1927). Areas investigated are shown black (modified after Behr et al., 1980 and Riding, 1974). *A* = Harz mountains; *B* = Franconian Forest; *C* = Vosges.

during the Hercynian mountain building. In this paper we present first palaeomagnetic data from the Hercynian mountain belt of central Europe covering the Devonian to Carboniferous time span. These data are compared with existing results from Great Britain (Briden et al., 1973; Duff, 1980), the Armorican Massif (Jones et al., 1979) and from the Central Massif (Edel et al., 1981).

Our sampling has concentrated on nonmetamorphic or very low-grade metamorphosed Palaeozoic rocks in the Franconian Forest, the Harz mountains and the Vosges (Fig. 1). These areas represent different tectonic domains and different sedimentary basins within the Hercynian framework and are separated by different suture zones as proposed by Burrett (1972), Dewey and Burke (1973), Laurent (1972) and Riding (1974). The sutures zones are more or less identical with the boundaries between the Rhenohercynian, the Saxothuringian and the Moldanubian (Kossmat, 1927) zones of Hercynian central Europe.

SAMPLING AND LABORATORY TECHNIQUES

Samples collected for the present study represent sediments and volcanics of Middle Devonian to Early Carboniferous age from the Rhenohercynian (Harz), the Saxothuringian (Franconian Forest) and the Moldanubian (Vosges) zones of the Hercynian "geosyncline". The sediments consist mainly of shales, cherts, greywackes and limestones. The volcanics are mainly diabases; they have been sampled in the Franconian Forest and in the Harz mountains.

The samples were cored with a gasoline-powered portable drill and oriented with a magnetic compass. Diabase outcrops were sampled only at sites where direct contact with sedimentary strata provided the possibility for a clear structural correction. At one site the layered structure of a pillow lava served as a reference for bedding correction.

All samples (2.54 cm diametre) were measured with a 3-axis cryogenic magnetometer interfaced to a minicomputer. This measuring configuration has a lower sensitivity limit of $6 \cdot 10^{-06}$ A/m. Thermal demagnetization was performed in air in a field-free furnace to maximum temperatures of 700°C. Alternating field demagnetization up to fields of 60 mT was carried out using a two-axis tumbler and a Helmholtz coil system for compensation of the earth's magnetic field. During the transport from the demagnetization device to the magnetometer the samples were kept in mu-metal shields. The cryogenic magnetometer is surrounded by a field-compensating Helmholtz coil so that remagnetization by the earth's magnetic field during laboratory treatment was kept at a minimum. The directions and intensities of the natural remanent magnetization (NRM) were plotted on non-coherent vector diagrams (modified after Zijderveld, 1967) in order to establish the characteristic magnetization components.

MAGNETIC PROPERTIES

Late Devonian to Early Carboniferous granites and multiply deformed migmatitic gneisses from the Black Forest do not possess suitable palaeomagnetic characteristics. On the other hand we have found non-metamorphic sediments and volcanics from the Hercynian realm to contain stable magnetization components which may be used successfully for a palaeomagnetic analysis. However, severe remagnetization problems have been encountered when studying the palaeomagnetism of Palaeozoic rocks in Europe and North America (Creer, 1967; McElhinny and Opdyke, 1973). The NRM of Old Red Sandstone sediments in Scotland has been interpreted (Chamalaun and Creer, 1964) to reflect both partly mixed and partly one component field directions even at the highest thermal demagnetization stages. Overprinting occurred during Permo-Carboniferous times and was ascribed to different formation processes of secondary magnetic oxides (Creer, 1968). Strong secondary magnetizations are also reported for Palaeozoic rocks from the Rhenic Shield (Heller and Kneidl, 1983; J.A. As, pers. commun., 1981) and the Franconian Forest (Kim and Soffel, 1982). Therefore, prior to interpreting NRM directions, the magnetic properties of the three main rock types investigated (limestone, greywacke, diabase) have to be discussed in some detail.

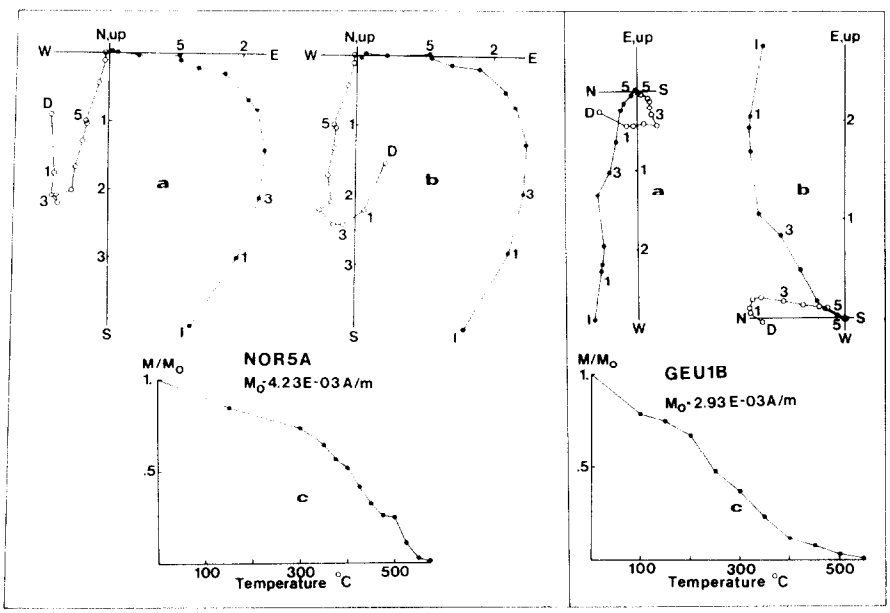


Fig. 2. Thermal demagnetization of red (sample NOR 5A) and white (sample GEU 1B) Devonian limestone from the Franconian Forest. Vector plot of the declination (solid circles) and inclination (open circles) components (a) before and (b) after bedding correction, (c) intensity versus temperature. Unit on vector plots: $1 \cdot 10^{-03}$ A/m.

The NRM intensities of the cephalopod bearing, partly reddish (Fig. 2; sample NOR 5A), partly white to grey (Fig. 2; sample GEU 1B) Middle to Upper Devonian *limestones* have an average intensity of $3.5 \cdot 10^{-03}$ A/m. Both samples contain a large present-day field component (subfigures 2a) which is removed at temperatures higher than 300°C. Above 400°C single components of both polarities remain with southerly declination and shallow positive inclination or northerly declination with negative inclination. The high temperature direction prior to structural correction of the white limestone clearly indicates that an eventual remagnetization during the Permo-Carboniferous (expected direction: declination = 200°, inclination = −25°) did not occur. This magnetization is therefore considered to be contemporaneous with the deposition of the rocks. Although high coercivity minerals contribute to high field magnetization in the red limestone variety, maximum unblocking temperatures do not exceed 575°C. We infer that the characteristic high-temperature component of both limestone types is carried by magnetite. The apparently higher stability of NRM intensity (subfigures 2c) of the red variety is caused by the contribution of pigmentary haematite with relatively low unblocking temperatures. This behaviour has also been observed in Mesozoic pelagic limestones (Scaglia Rossa) from the South Tethian realm (Channell et al., 1982).

The Devonian and Carboniferous *greywackes* have slightly lower NRM intensities than the limestones. The NRM mean intensity is around $1.0 \cdot 10^{-03}$ A/m. The acquisition curves of isothermal remanent magnetization (IRM) show that the magnetic mineralogy is dominated by high coercivity haematite (Fig. 3g). Progressive thermal and AF demagnetization removes secondary components of unknown origin A stable magnetization vector is found at temperatures higher than 400°C which has the same direction as the stable endpoint obtained during AF cleaning between 35 mT and 55 mT. The evidence from a positive fold test in the Harz mountains (see Fig. 5) and the directional stability support the characteristic directions of these greywackes to represent an early, probably synsedimentary, imprint of magnetization. The maximum unblocking temperatures, however, cannot be determined because the greywackes undergo major magnetomineralogical changes above 525°C and build up highly viscous remanences (due to new formation of magnetite(?)) obscuring the original NRM signal.

Except for a few spilitized diabases (albite–chlorite paragenesis: Wirth, 1978) which are not considered in this study, the *volcanics* are usually strongly magnetized with an initial NRM mean value at $5 \cdot 10^{-01}$ A/m. The IRM acquisition curves (Fig. 4d) indicate a low coercivity mineral which was identified under the microscope as (Ti?-)magnetite. The amount of sulphides observed optically is very small. Major contributions of such mineral phases to the magnetic properties are not indicated by the NRM intensity decay during thermal demagnetization (Figs. 4c: no inflexion between 300°C and 400°C), and maximum blocking temperatures are reached around 575°C. Again strong secondary magnetization components produced by the influence of the present earth field are eliminated at 500°C. At higher temperatures

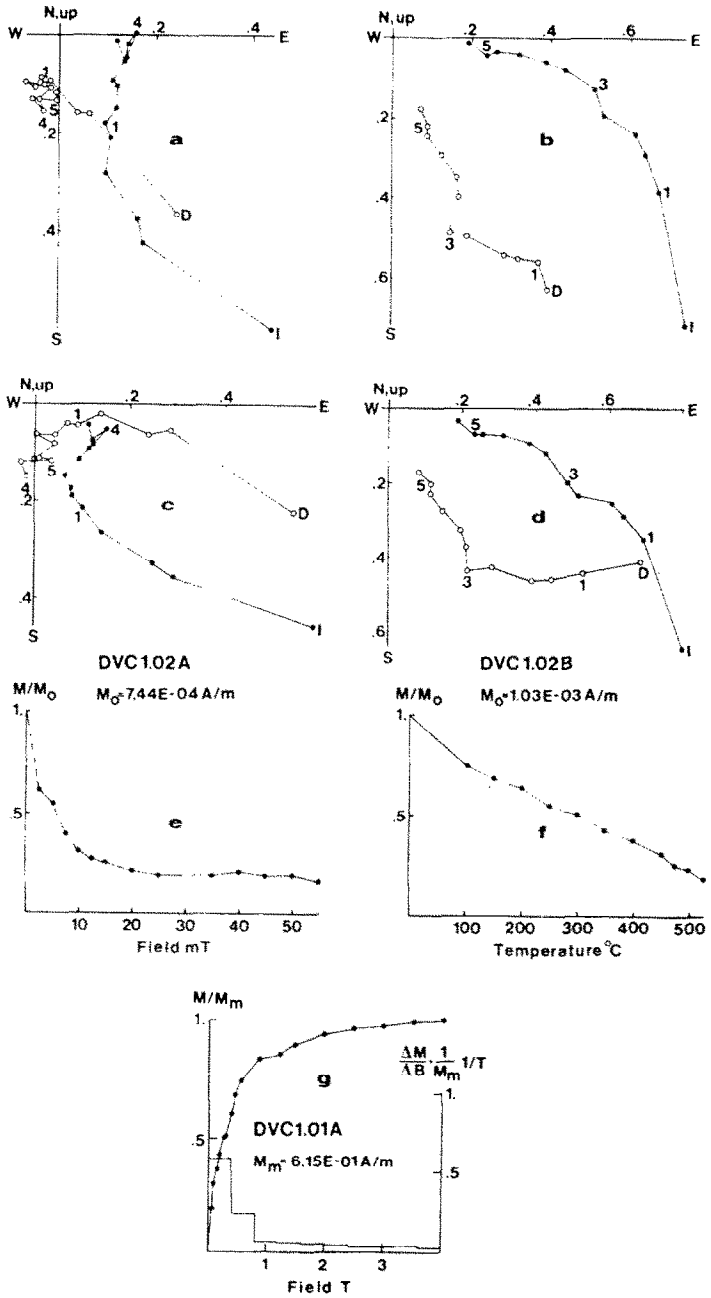


Fig. 3. Thermal (b, d, f) and AF-demagnetization (a, c, e) of an Upper Devonian greywacke from the Vosges: (a, b) before, (c, d) after bedding correction, (e, f) NRM intensities versus temperature respectively AC-field. (g) IRM acquisition with additional coercivity spectrum. Unit on vector plot: $1 \cdot 10^{-03}$ A/m.

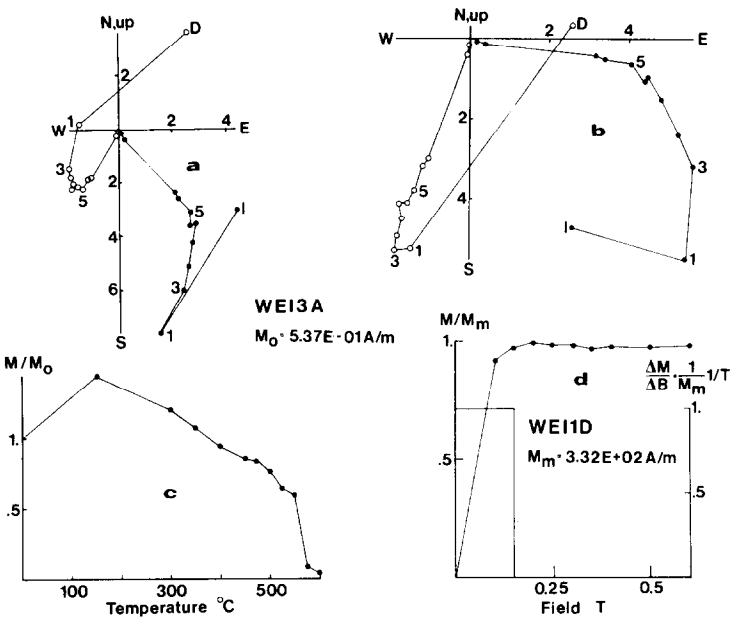


Fig. 4. Thermal demagnetization of Devonian diabase from the Franconian Forest: (a) before, (b) after bedding correction, (c) NRM intensity vs. temperature. (d) IRM acquisition with additional coercivity spectrum. Unit on vector plot: $1 \cdot 10^{-01}$ A/m.

a single NRM component remains (Figs. 4a, b). Before dip correction (Fig. 4a) this component is not related to a Permo-Carboniferous field direction, but has southerly declination and downward inclination. Remagnetization seems to be unlikely in these fresh diabase rocks and the characteristic direction may still preserve the Devonian field.

GEOLOGICAL SETTING AND PALAEOMAGNETIC RESULTS

Vosges

In the Moldanubian part of the southern Vosges three sites were sampled 30 km north of Belfort in the Palaeozoic Markstein series. This lithological unit consists mainly of Upper Devonian and Lower Carboniferous greywackes and intercalated shales in marginal to basinal facies (R. Maas, pers. commun., 1981). The tectonic style of this area is characterized by strike-slip faults and small scale overthrusts of probable Late Devonian to Early Carboniferous age.

The characteristic NRM mean directions of the Upper Devonian greywackes yield a mean palaeopole at −24.7°N and 10.7°E. The Lower Carboniferous pole for this area (based on two sites) was determined at −40.9°N, 344.2°E after tectonic

TABLE I

Mean directions and pole positions of the areas investigated *

Area and lithology	Age	N	n	Decl/Incl before structural correction	α_{95}	VGP before structural correction	Decl/Incl after structural correction	α_{95}	VGP after structural correction
Vosges									
Greywacke	Cl	2	16	022.7/+14.7	06.5	−45.1 N, 334.2 E	017.4/+02.4	06.5	−40.9 N, 344.2 E
Greywacke	Du	1	12	182.5/+07.8	09.4	−37.1 N, 007.4 E	182.5/+22.9	09.1	−24.7 N, 010.7 E
Harz									
Greywacke	Cl	2	32	140.1/+43.0	18.6	−05.6 N, 046.6 E	173.1/−09.0	04.9	−42.4 N, 020.2 E
Diabase	Du	1	6	174.4/+28.6	08.9	−22.8 N, 016.7 E	173.3/+16.8	08.9	−29.3 N, 018.3 E
Limestone	Dm	1	11	165.0/+51.7	12.1	−04.8 N, 023.5 E	163.6/+31.7	12.1	−19.6 N, 027.4 E
Franconian Forest									
Chert	Cl	1	21	202.2/+17.8	10.2	−27.7 N, 346.2 E	201.7/−07.8	10.2	−40.4 N, 342.5 E
Diabase	Dm−u	5	43	198.6/+42.5	07.0	−13.3 N, 354.4 E	176.9/+28.9	07.0	−24.2 N, 015.1 E
Limestone	Dm	2	13	206.4/+11.5	11.5	−16.5 N, 354.8 E	190.6/+22.2	11.5	−27.5 N, 000.1 E

* *Cl* = Lower Carboniferous, *Dm* = Middle Devonian, *Du* = Upper Devonian, *Dm-u* = combined Middle and Upper Devonian, *N* = number of sites, *n* = number of samples used to calculate the NRM mean directions; α_{95} confidence angle at the 95% probability level.

dip correction (Table I). The between site scatter of the palaeomagnetic directions is low and apparently was not affected by the tectonic features mentioned above.

Harz

Limestones of Middle to Late Devonian age (Iberg reef), Upper Devonian diabase and Lower Carboniferous greywackes representing the most northerly outcrops of the Hercynian mountain belt in Germany (Rhenohercynian zone after Kossmat, 1927), have been sampled in the Harz mountains. Deposition in this area is documented from Silurian to Early Carboniferous times. During the Devonian and Carboniferous deposition of shales, cherts, sandstones and limestones predominated. In the Middle Devonian (Givetian) swells and basins developed in the Harz geosyncline (Alberti and Walliser, 1977), the swells being marked by the effusion of diabases with huge reef complexes on top and the basins being filled with fine-grained detrital material (shales). Alberti and Walliser (1977) consider the oldest greywackes within the Harz mountains to be of Late Devonian age. However, most of the greywackes have been deposited in a rapidly subsiding basin during the beginning of the main Hercynian orogenic event in Viséan times. The tectonic style of the Harz is dominated by NE–SW-striking folds.

Some Upper Devonian and Lower Carboniferous diabases are altered (spilitization), and according to the analysis of their magnetization directions they were remagnetized in the Permian or Tertiary. They require further investigation and are not discussed in this paper. The cleaned mean NRM direction of an unaltered, fresh Devonian diabase gives a virtual geomagnetic pole (VGP) position of $-29.3°$N and 18.3°E. Although very weakly magnetized, the Devonian limestones reveal a reasonably well-defined characteristic mean direction. The resulting palaeopole position is given by $-19.6°$N and 27.4°E (Table I). A fold test within a slightly north-vergent fold could be carried out in Lower Carboniferous greywackes (Fig. 5). The fold test proved to be positive at the 95%-probability level (McElhinny, 1964). Thus the characteristic remanence predates the folding which occurred shortly after deposition of the sediments (Alberti and Walliser, 1977).

Franconian Forest

The Franconian Forest in northeastern Bavaria is situated northwest of the Münchberg gneiss-massif and is part of the Saxothuringian zone. It consists of a large variety of sedimentary facies, which are grouped into two major realms, the Bavarian and the Thuringian facies (Wurm, 1927). The Bavarian facies is restricted to the vicinity of the Münchberg gneiss-massif, whereas the Thuringian facies covers most of the Franconian Forest. In both types of facies sedimentation is documented from Cambrian to Carboniferous times (Behr et al., 1980).

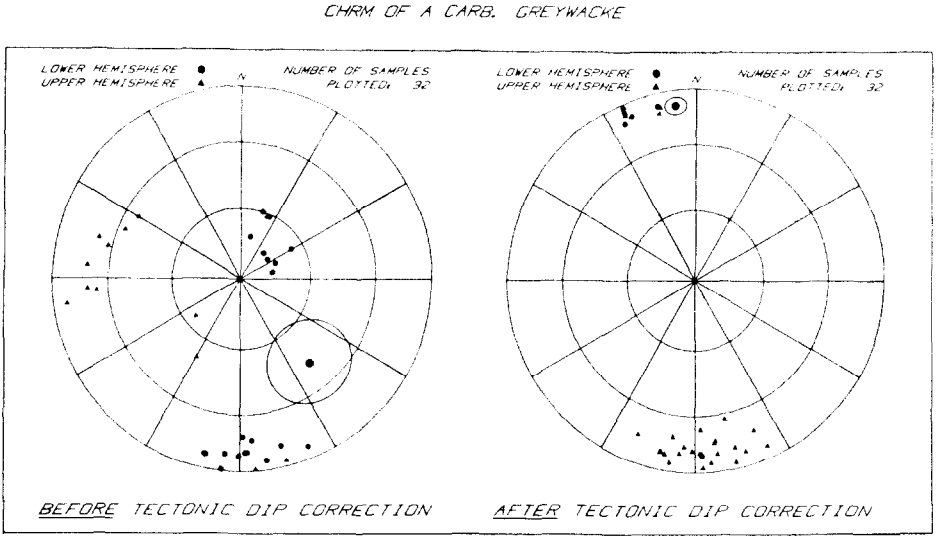


Fig. 5. Equal-area projection of characteristic NRM-directions of a Lower Carboniferous greywacke (Harz mountains).

A considerable amount of volcanic and tuffaceous rocks such as diabases and pyroclastics of Givetian to Frasnian age (Middle to Late Devonian) are found in the Thuringian facies. The volcanics display tholeiitic to alkaline character (Wirth, 1978), and recent studies of Brand (1979) show that they have been subjected only to low-grade regional metamorphism with temperatures ranging between 250° and 450°C. These volcanics are associated with conglomerates, tuffaceous greywackes and condensed shallow-water type cephalopod limestones. The Bavarian facies is represented by cherts, greywackes and some limestones. In the Carboniferous a flysch-type sedimentation (greywackes, shales and conglomerates) predominated in both the Thuringian and the Bavarian facies.

NE–SW trending fold axes and younger NW–SE striking fault planes control the tectonic framework of the area. In the vicinity of the Münchberg gneiss-massif a more complicated tectonic pattern consisting of nappe structures and large-scale overthrusts has been demonstrated for the Bavarian facies, probably caused by the uplift of the gneiss-massif during Late Devonian to Early Carboniferous times (Behr et al., 1980). Therefore palaeomagnetic sampling was restricted to rocks of the tectonically less disturbed and autochthonous Thuringian facies.

Sampling concentrated mainly on the well exposed diabases and to a smaller extent on cephalopod limestones and cherts. These rock types yielded well grouped characteristic NRM mean directions. The Middle to Upper Devonian diabases give a well-defined regional mean pole position after tectonic dip correction at −24.2°N and 15.1°E. The Upper Devonian limestones being uncertain in tectonic position

(W. Franke, pers. commun., 1982) yield a pole situated at −27.5°N and 0.1°E; the Carboniferous cherts (sampled at one site) yield a VGP position at −40.4°N and 342.5°E (Table I).

DISCUSSION

The magnetic behaviour of the rocks investigated including the positive greywacke test suggests that the characteristic NRM directions have been acquired during or shortly after rock formation. Although an overprint of probable Permian and/or Tertiary age has been observed at some localities which are not reported here (e.g. the Sauerland and Hunsrück mountains, and a few sites in the Harz), the

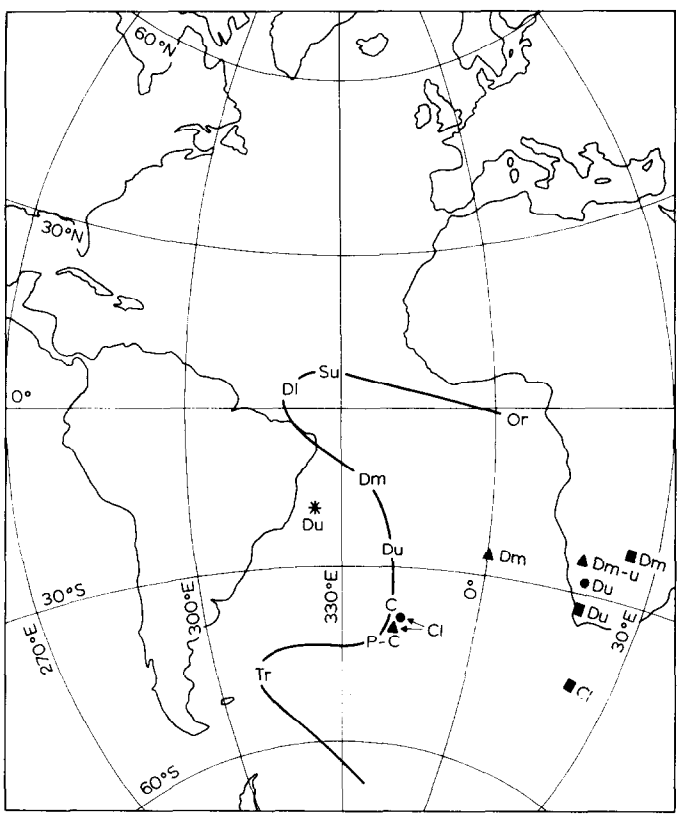


Fig. 6. Middle Devonian to Lower Carboniferous poles from the Vosges (dots), the Franconian Forest (triangles) and the Harz mountains (squares) combined with the Palaeozoic polar wander path for Palaeozoic Europe according to Briden et al. (1973), and Duff (1980). The Upper Devonian palaeopole for Britanny (Jones et al., 1979) is denoted by a star.
Or = Ordovician, *Su* = Upper Silurian, *Dl* = Lower Devonian, *Dm* = Middle Devonian,. *Du* = Upper Devonian, *Dm-u* = combined Middle and Upper Devonian, *Cl* = Lower Carboniferous, *P-C* = Permo-Carboniferous and *Tr* = Triassic.

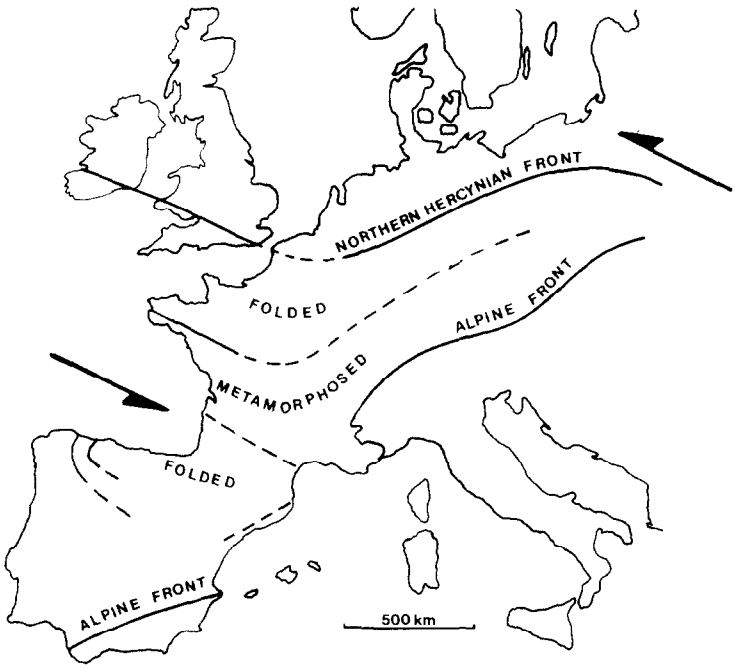


Fig. 7. Sketch map of the Hercynides (modified after Riding, 1974) as a large megashear belt. The supposed shear directions are indicated by arrows.

palaeomagnetic poles presented in this paper predate the main Variscan folding events in central Europe. They can be used for an analysis of the Palaeozoic geomagnetic field and for a tentative tectonic interpretation of the Hercynian belt.

For such an interpretation the central European data are compared with the apparent polar wander path (APWP) of northern Europe derived for the Palaeozoic by the extensive studies of Briden et al. (1973) and Duff (1980). This polar wander curve has been plotted in Fig. 6 together with the pole positions listed in Table I. The palaeolatitude of the central European areas, as derived from the Devonian and Lower Carboniferous regional mean inclinations (Table I), is quite similar to that observed in England and Britanny. This confirms the argument of Jones et al. (1979) that a wide north–south separation of the northern European Foreland and Hercynian Europe with formation of a large intervening Palaeozoic ocean as proposed by Morel and Irving (1978) is not required by the palaeomagnetic results.

Two Lower Carboniferous pole positions from the Franconian Forest and the Vosges are consistent with the extra-Hercynian apparent polar wander path, thereby indicating no significant relative movement between these regions (Fig. 6). The Carboniferous pole from the Harz mountains, however, differs in declination by an amount of about 20°. This pole is based on data from a strongly deformed site which yields a positive fold test (Fig. 5). Due to the deformation which may be

related to small-scale nappe emplacement (Alberti and Walliser, 1977), it is not clear at present whether this NRM mean direction has local significance or reflects regional tectonic displacement.

The final pole positions of the Devonian rocks of the three regions investigated cluster rather tightly, indicating that the pole positions are not affected by local post-depositional tectonics. They differ, however, significantly in longitude from the European APWP. An anti-clockwise rotation of Hercynian central Europe by about 30° fits both data sets. Since in contrast to observations of Edel et al. (1981) our Carboniferous data are consistent between foreland and mountain belt, the rotational movement must have occurred around the Devonian/Carboniferous boundary, perhaps during a tectonic event (Sudetic?) which preceeded the main flysch sedimentation in the Early Carboniferous. Also, our data suggest no large-scale convergent motion between northern and central Europe during the evolution of the Hercynian geosyncline. Considerable lateral displacement or rotation of continental segments may therefore have played an important role (Lorenz, 1976, Edel et al., 1981). Since typical continental collision signatures are lacking, the Hercynides may constitute a large-scale shear belt as proposed by Arthaud and Matte (1977). We cannot recognize differential motions between the tectonic units investigated. Therefore we suggest that large megashears with general east–west orientation caused relative anticlockwise rotation of a central European block by about 30° of arc with respect to the northern Hercynian foreland as depicted in Fig. 7. The internal amount of rotation within the Hercynian mountain belt between Armorica (Jones et al., 1979) and central Europe is even larger and of the order of 45 degrees of arc (see distribution of poles in Fig. 6).

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